

Qualitative Disclosure as a Tax Enforcement Mechanism: Evidence from the U.K. Tax Strategy Disclosure Requirement

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Abstract: In 2016, the U.K. passed a regulation that requires large businesses to publicly disclose their tax strategy. The U.K. regulator expects these qualitative disclosures to attract public scrutiny of firms' tax practices, thereby pressuring firms to reduce tax avoidance. This study examines whether the U.K. tax strategy disclosure requirement has achieved this objective. Using a difference-in-differences design and a sample of U.K. publicly traded firms, I find evidence that is most compatible with the regulation not having a significant impact on firms' tax avoidance. Inferences are similar when I focus on subsamples that are most likely to exhibit the intended behavioral changes using a series of cross-sectional tests within treated firms. Thus, the collective evidence is largely inconsistent with the regulation successfully curbing tax avoidance, which should inform regulators worldwide as they consider implementing similar disclosure regulations to combat corporate tax avoidance.

Keywords: Tax strategy disclosure, public scrutiny, tax avoidance

Data Availability: Data in this study are obtained from public sources as identified in the paper.

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Abstract: In 2016, the U.K. passed a regulation that requires large businesses to publicly disclose their tax strategy. The U.K. regulator expects these qualitative disclosures to attract public scrutiny of firms' tax practices, thereby pressuring firms to reduce tax avoidance. This study examines whether the U.K. tax strategy disclosure requirement has achieved this objective. Using a difference-in-differences design and a sample of U.K. publicly traded firms, I find evidence that is most compatible with the regulation not having a significant impact on firms' tax avoidance. Inferences are similar when I focus on subsamples that are most likely to exhibit the intended behavioral changes using a series of cross-sectional tests within treated firms. Thus, the collective evidence is largely inconsistent with the regulation successfully curbing tax avoidance, which should inform regulators worldwide as they consider implementing similar disclosure regulations to combat corporate tax avoidance.

I. INTRODUCTION

The global corporate tax environment has shifted dramatically in the past decade to focus more acutely on the direct impact of corporate tax revenue on social welfare and economic equity (e.g., PwC 2022; Elliott 2022). From the U.K. consumer boycott following the negative publicity surrounding Starbucks' tax practices to the European Commission's high-profile crackdown on Ireland's favorable tax treatment of Apple, a growing set of stakeholders are pressuring businesses to pay their "fair share" of taxes. Consistent with this trend, in 2016, a new U.K. regulation requiring large businesses to publish their tax strategy received legislative approval. The regulator expected these disclosures to enable greater public scrutiny over firms' approach to tax, which will in turn curb tax avoidance (HMRC 2015b, 11). This study examines whether the U.K. tax strategy disclosure requirement has achieved the objective of reducing corporate tax avoidance.

Tax authorities worldwide have implemented various private or public tax disclosure initiatives to combat corporate tax avoidance (Hoopes, Robinson, and Slemrod 2024). Compared to other existing public tax disclosure initiatives, the U.K. tax strategy disclosure requirement is unique in several aspects. First, the U.K. regulation only mandates *qualitative* information about firms' tax strategy, including 1) approach to tax risk management, 2) attitude toward tax planning, 3) tax risk appetite, and 4) approach to interacting with Her Majesty's Revenue and Customs (HMRC), the U.K. tax authority. The qualitative nature of the disclosure constitutes a stark contrast to other public tax disclosure regimes that seek *quantitative* information such as the amount of tax payments to combat tax avoidance. Second, the U.K. regulation differs from public disclosure of tax-related information in the financial statements mandated by accounting standard setters or capital market regulators (e.g., the disclosure of unrecognized tax benefits in the U.S.). While the latter disclosure is intended to provide investors with value-relevant information, such as the

impact of tax on future after-tax cash flows (Hoopes et al. 2024), the U.K. tax strategy disclosure is not designed for this purpose. Instead, the regulator expects tax strategy disclosures to attract public scrutiny of firms' tax practices, making the public the intended audience of these disclosures. Finally, the regulation does not require that firms include their tax strategy disclosures in the audited annual reports. Instead, firms can choose where to publish the disclosure. Given these unique features, the U.K. tax strategy disclosure requirement represents a novel public disclosure regulation to curb tax avoidance.

The U.K. regulation mandates disclosure of *tax strategy* rather than specific tax planning activities because the regulator believes that the level of formalization and the content of a firm's tax strategy are clear indicators of its stance on tax avoidance (HMRC 2015b). To allow for more informative disclosures, the regulation adopts a principles-based disclosure approach with a minimum disclosure threshold. Specifically, it only requires that firms disclose some information about each mandatory disclosure component to ensure consistency in the scope and comparability of disclosures across firms. However, firms can choose the level of detail and the specific content of the disclosures. The regulator expects that offering flexibility in disclosure "will minimise the risk of businesses publishing sanitised 'generic' tax strategies" (HMRC 2015c, 11). This view is consistent with recent research, which finds that firms' tax avoidance is associated with tax-related disclosure choices and characteristics (e.g., Inger, Meckfessel, Zhou, and Fan 2018; Adams, Demers, and Klassen 2022). Thus, disclosures could reveal firm "type" under a principles-based approach, suggesting that tax strategy disclosures will be useful to the public in identifying and scrutinizing firms that engage in more tax avoidance. The regulation also requires the disclosures to be public because "reputational concerns can influence the attitudes of large businesses and encourage them to pursue less aggressive tax planning arrangements" (HMRC 2015b, 12). Surveys

and anecdotal evidence suggest that the U.K. has a strong and negative public sentiment toward corporations not paying their “fair share” of taxes (Dyrenge, Hoopes, and Wilde 2016; Maynard and Watt 2016; Pegg 2017). Thus, the overall corporate tax environment in the U.K. should assist the regulation in leveraging public scrutiny to combat tax avoidance.

Although the regulator expects the regulation to curb tax avoidance, there are a few concerns about this expectation. First, a principles-based disclosure approach requires costly enforcement and could be counter-productive if it allows too much discretion (Leuz 2010). If most firms provide minimal information or use extensive boilerplate language to simply ensure compliance, tax strategy disclosures will not be useful in identifying tax avoiders, which is a necessary condition for public scrutiny to pressure for behavioral changes. Second, existing research provides mixed evidence on whether and to what extent negative public perception can change tax avoidance. In fact, most findings are consistent with public scrutiny not having enough “teeth” (e.g., Gallemore, Maydew, and Thornock 2014; Hasegawa, Hoopes, Ishida, and Slemrod 2013; Hoopes, Robinson, and Slemrod 2018; Chen, Schuchard, and Stomberg 2019). Thus, even in the U.K., where the overall corporate tax environment is favorable to the regulation in achieving its objective, it remains unclear whether a disclosure regulation is able to leverage public scrutiny to change firms’ tax practices. Finally, allowing firms to choose where to publish their tax strategy makes it difficult for the public to locate these disclosures. As information awareness and acquisition costs increase, the level of public scrutiny that these disclosures can attract will decrease, and the potential impact on firm behavior will weaken (Blankespoor, deHaan, Wertz, and Zhu 2019; Maines and McDaniel 2000). Thus, whether the U.K. tax strategy disclosure requirement will achieve its objective of curbing tax avoidance is an empirical question.

To examine the research question, I construct a sample of U.K.-incorporated firms that are

publicly listed on the London Stock Exchange from 2010 to 2019 with the necessary data to create regression variables. I focus on U.K.-incorporated firms because their British identity makes them more likely and willing to change tax behavior in response to public scrutiny than non-U.K. firms (HMRC 2015b). This sample selection choice also holds the tax enforcement environment and country institutions constant across sample firms. The regulation applies to U.K. firms, groups, or subgroups with total assets above £2 billion or turnover greater than £200 million (Finance Act 2016). Accordingly, I classify firms as treated if they meet at least one disclosure threshold. I measure tax avoidance with one-year book and cash effective tax rates (ETRs). Using a difference-in-differences (DiD) research design, I estimate statistically insignificant changes in tax avoidance for treated firms relative to control firms post-regulation. I confirm that the 95 percent confidence intervals for the coefficient estimates of interest include zero in all specifications (Cready et al. 2022). The upper bound estimates, which capture the largest estimated effect in the predicted direction if the regulation is effective, correspond to an average 12 (five) percent reduction in book (cash) ETR relative to the sample mean (Cready et al. 2022). Taken together, these findings are most compatible with no differential changes in tax avoidance between treated and control firms, which is inconsistent with the objective of the regulation.

To complement the DiD design, I conduct several cross-sectional tests within treated firms. These tests minimize the concern that changes in tax avoidance among the control firms could explain the primary findings. More importantly, by focusing on firms that are most likely to exhibit the intended behavioral responses, these tests provide additional evidence on the effect of the regulation, even if such an effect only exists in a subsample of treated firms. I first partition treated firms based on the ownership of valuable brands. Austin and Wilson (2017) show that firms with valuable brands are more sensitive to public perceptions of their tax practices due to the potentially

negative impact on brand value. Thus, all else equal, I expect the effect of the regulation to be stronger among firms with valuable brands. Next, I partition firms based on the existence of tax haven operations. Because the U.K. public negatively perceives tax haven operations (Dyrenge et al. 2016), I expect that firms with tax haven subsidiaries are more likely to attract public scrutiny and change tax avoidance. In both cross-sectional tests, I find evidence that is most compatible with no differential changes in tax avoidance, even among these subsamples of treated firms that are most likely to exhibit the intended behavioral responses. These findings complement the DiD analysis and are inconsistent with the regulation successfully curbing tax avoidance.

I conduct three supplemental tests, including implementing a regression discontinuity (RD) design, examining changes in U.K. tax avoidance, and addressing the potential impact of the Country-by-Country (CbC) reporting requirement. I find qualitatively similar results in all tests, which corroborates the primary evidence and should alleviate the concern of an alternative explanation. Although I cannot conclude definitively that firms subject to the U.K. tax strategy disclosure requirement did not change tax avoidance, the collective evidence is most compatible with the regulation not being effective in curbing tax avoidance.

This study has important policy implications. The U.K. tax strategy disclosure requirement is the first *mandatory, qualitative* public tax disclosure regulation that solely focuses on firms' tax practices. Despite the regulator's belief that this regulation will be effective in curbing tax avoidance, my findings are inconsistent with such an expectation and most compatible with the regulation—at least in its current form—not having a significant impact on firms' tax practices. From a policy perspective, finding that a policy does not appear to work as intended should be equally valuable as evidence supporting the effectiveness of such policy.

In recent years, the demand for public, qualitative tax disclosure has increased around the

world, and this U.K. regulation has attracted the attention of regulators in other countries. For example, the Australian Taxation Office (ATO) incorporated tax strategy disclosure into its Voluntary Tax Transparency Code in 2018 (ATO 2018), and publicly traded firms are expected to publish similar tax disclosures in Spain and Denmark (KPMG 2021). In 2019, the Global Reporting Initiative (GRI) standards—a widely used framework for sustainability reporting—introduced a tax topic, which requires tax strategy disclosures similar to those mandated by the U.K. regulation (GRI 2019). These global developments in disclosure initiatives indicate a growing interest in public, qualitative tax disclosure. By leveraging the U.K. tax strategy disclosure requirement, which is the only disclosure initiative of such kind that has been implemented, I provide empirical evidence suggesting that these disclosure regimes are also unlikely to constrain tax avoidance if they follow the approach implemented in the U.K. Thus, the findings of this study are relevant to regulators who are interested in implementing a similar disclosure initiative.

This study also contributes to the tax disclosure literature by examining the effectiveness of a novel public disclosure regulation that differs from existing public tax disclosure regimes. While recent studies find some promising evidence that private CbC reporting is effective in reducing tax avoidance and has real effects on firms' investment decisions (e.g., Joshi 2020; De Simone and Olbert 2022), my findings appear to suggest that a public, qualitative tax disclosure initiative may not have enough “teeth” in changing tax avoidance, even though the design features are appealing to the regulator. The collective evidence of this study and research on other tax disclosure initiatives should assist regulators in evaluating different disclosure initiatives to avoid a potential “crowding out” effect when introducing multiple disclosure regulations.

Finally, this study complements two other concurrent studies examining different aspects of the U.K. regulation. Belnap (2023) conducts a field experiment, where he initiates a public

campaign against U.S. multinational firms that are subject to this U.K. regulation. He finds that after being exposed to public naming-and-shaming by reputable media outlets, firms that were previously non-compliant with the regulation responded by providing required disclosures, and those that were compliant improved disclosure quality. These findings suggest that public scrutiny, if leveraged to its full potential, can induce changes in disclosures. Bilicka, Casi, Seregni, and Stage (2022), a concurrent working paper, documents an increase in tax disclosure in annual reports after the regulation became effective, suggesting a potential spillover benefit. Consistent with this study, they do not find that firms reduce tax avoidance post-regulation. Given that Bilicka et al. (2022) focus on disclosures in annual reports, their findings indicate that the saliency of the disclosure (i.e., on corporate websites or in annual reports) may not be the primary explanation for the lack of intended behavioral changes. The fact that two independently designed studies reach similar conclusions should lend creditability to each other and jointly, they suggest that the U.K. regulation likely has little impact on firms' tax avoidance. While Bilicka et al. (2022) primarily examine how this U.K. regulation influences *other* tax disclosures, and Belnap (2023) studies the extent of *compliance* with the disclosure among *U.S.* firms, this study focuses on the effectiveness of the regulation in achieving its objective among U.K. firms using various research designs to provide corroborating evidence. Collectively, these three studies provide a comprehensive picture of the effect of this novel qualitative tax disclosure initiative.

II. BACKGROUND AND HYPOTHESIS DEVELOPMENT

Overview of the U.K. Tax Strategy Disclosure Requirement

The U.K. tax strategy disclosure requirement was first proposed by the HMRC in July 2015, when the initial draft policy was published for public consultation. After soliciting feedback from various stakeholder groups, the HMRC revised the draft policy and submitted it for legislative

approval. On September 15, 2016, the U.K. tax strategy disclosure requirement obtained Royal Assent as Schedule 19 of the U.K. Finance Act 2016 and thus, became law (U.K. Parliament 2016). The U.K. regulation requires qualifying entities to publicly disclose tax strategy on an annual basis, effective for fiscal years beginning after September 15, 2016. Qualifying entities include any U.K. firms, groups, or subgroups with total assets over £2 billion or turnover (i.e., similar to sales under U.S. GAAP) above £200 million in the preceding fiscal year. Qualifying entities also include multinational firms with a U.K. presence and global turnover above €750 million (Finance Act 2016). The regulation mandates four disclosure components: 1) tax risk management, 2) attitude toward tax planning, 3) tax risk appetite, and 4) the firm's approach to dealing with the HMRC (Finance Act 2016). To comply, firms must discuss all four components in relation to U.K. taxation but are *not* required to disclose quantitative information such as the amount of tax payments. The regulation does not require these disclosures to be audited, and firms can choose where to publish their tax strategy (e.g., corporate website) as long as the disclosures are publicly accessible free of charge (HMRC 2016a). Appendix A provides two disclosure examples.

The U.K. tax strategy disclosure requirement imposes a £7,500 initial penalty on nondisclosure, incomplete disclosure, or failure to keep the disclosure publicly accessible. Additional charges apply if a non-compliant firm fails to take corrective action (HMRC 2016a). In addition to financial penalties, the HMRC considers a firm's compliance with this regulation and the extent to which the disclosed tax strategies align with the firm's tax practice during the internal business risk review process (HMRC 2016b). This review process allows the HMRC to classify business taxpayers into different risk levels using both "hard" (e.g., size, nature of the business operations) and "soft" information (e.g., internal governance, prior interactions with the HMRC). Thus, untruthful disclosure is unlikely because it will negatively affect a firm's tax risk

rating—a key factor that guides the HMRC’s approach to dealing with the firm.

U.K. Regulation vs. Existing Public Tax Disclosure Initiatives

Tax authorities worldwide use both private and public disclosure initiatives to combat tax avoidance. Most existing private tax disclosure regimes require firms to provide information solely to the tax authority to help reduce information asymmetry between taxpayers and the tax authority. A notable example is the private CbC reporting framework under Action 13 of the Base Erosion and Profit Shifting (BEPS) project, which requires firms to disclose economic profits and global tax payments by jurisdiction to the tax authority.¹ Because the information required under private disclosure regulations is usually proprietary in nature and otherwise unavailable to the tax authority, changes in firms’ tax behavior, if any, are likely attributable to the deterrent effect of future tax enforcement or actual tax enforcement activities (Hoopes et al. 2024).

In contrast, the U.K. tax strategy disclosure requirement is a mandatory, *public* tax disclosure initiative. Compared to existing mandatory, public tax disclosure regimes, this U.K. regulation is unique in several aspects.² First, the U.K. regulation focuses on firms’ tax strategy rather than specific tax planning activities, and it purposefully excludes numeric information from being a mandatory disclosure component (HMRC 2015a). Thus, the disclosure content under the U.K. regulation is *qualitative* in nature, which differentiates this regulation from existing mandatory, public tax disclosure regimes that seek *quantitative* information. An example of the latter is the Australian tax return disclosure regulation, which requires the ATO to publish select tax return line items (e.g., taxable income and tax payable) annually with identifying information

¹ The U.K. adopted the CbC reporting requirement in 2017—the first year when the U.K. tax strategy disclosure requirement became effective. I address the potential impact of the CbC regime in a supplemental test.

² The literature on tax disclosure is extensive. Thus, I limit the discussion in section II to *public* tax disclosure regimes that are most relevant to the U.K. regulation. Readers can refer to Hoopes et al.(2024) for a comprehensive review of the tax disclosure regulations.

for a subsample of firms (ATO 2018).

Second, the U.K. tax strategy disclosure requirement was proposed by the HMRC—the U.K. tax authority, which distinguishes it from public disclosure of tax-related information in financial statements initiated by accounting standard setters or capital market regulators (e.g., disclosure of unrecognized tax benefits in the U.S.). Whereas the goal of financial statement disclosure is to provide value-relevant information to investors (Hoopes et al. 2024), the U.K. tax strategy disclosure does not serve a similar purpose.³ In addition, the U.K. regulation is not primarily designed to provide new information about firms' tax avoidance activities to the tax authority. Instead, the general public is the target audience, and the regulator aims to leverage public scrutiny as the primary force to constrain tax avoidance.⁴

Given these unique features, the U.K. tax strategy disclosure requirement represents a novel public disclosure regulation to curb tax avoidance. In fact, this regulation is the first to *mandate* disclosure of *qualitative* information that is entirely about firms' tax practices. Since its implementation in 2017, there has been a growing demand for public, qualitative disclosure around the world. For example, Australia, Denmark, and Spain have introduced disclosure initiatives

³ Graham et al. (2012) review research on public disclosure of tax-related information in financial statements.

⁴ Research shows that tax authorities use public disclosure of tax-related information in financial statements to assist tax enforcement (Bozanic, Hoopes, Thornock, and Williams 2017; Hope, Ma, and Thomas 2013; Chi, Persson, Shevlin, and Urcan 2023; Fox and Wilson 2023). Because the objectives of financial reporting differ from those of tax reporting, such differences make the financial statement disclosures complementary to the private disclosures to the tax authority. However, the U.K. tax strategy disclosure is unlikely to serve a similar purpose because this regulation is proposed by the HMRC—the U.K. tax authority. More importantly, the HMRC has alternative information sources about firms' tax avoidance. For example, firms subject to this U.K. regulation interact directly and frequently with the HMRC on tax compliance and controversy issues throughout the year. They are also subject to other stringent private disclosure rules, a notable example of which is the mandatory Disclosure of Tax Avoidance Schemes (DOTAS). Under DOTAS, firms or anyone involved in a tax avoidance scheme must report such activity to the HMRC, who then uses the disclosure to enforce compliance and even amend tax laws to close loopholes. Given these alternative information sources, tax strategy disclosures are unlikely to provide the HMRC with new information about firms' tax avoidance. Discussions with a U.K. public accounting practitioner confirm this statement. For these reasons, tax authority scrutiny is unlikely to be a key mechanism that changes tax avoidance in this setting. However, the latent effect of such scrutiny is difficult to observe. Even if such scrutiny exists, it does not appear to manifest in differential tax avoidance given the primary findings of this study.

similar to this U.K. regulation (ATO 2018; KPMG 2021). In 2019, the GRI standards incorporated a new tax topic into its widely used framework for sustainability reporting. Effective in 2021, firms adopting the GRI standards will disclose their tax strategy, similar to what is required by the U.K. regulation, under the tax topic (GRI 2019). These recent developments in global disclosure initiatives indicate increasing interest in mandatory, public disclosure related to tax issues. As a forerunner of these new initiatives, the U.K. regulation provides a setting to understand the effectiveness of this novel disclosure regulation in combating tax avoidance.

Hypothesis Development

Predicted Effects of the U.K. Regulation: Regulator's Perspective

The goal of the U.K. tax strategy disclosure requirement is to “drive behaviour changes around tax planning and therefore enhance tax compliance” (HMRC 2015a). The regulator expects public disclosures of tax strategy to achieve this objective for two primary reasons: 1) disclosures will provide useful information to the public in scrutinizing firms’ tax practices, and 2) public scrutiny will have a meaningful impact such that firms change behavior.

The HMRC’s internal research finds that “the degree of codification and content of a tax strategy” are “clear indicators for aggressiveness in tax planning” (HMRC 2015a, 21), and firms with greater risk appetite tend to focus on minimizing tax payments to be no more than what is legally required (HMRC 2015b). These findings indicate that a firm’s tax strategy is inherently related to its stance on tax avoidance. Thus, the regulator considers tax strategy disclosures useful in understanding a firm’s tax practice. The regulation also adopts a principles-based disclosure approach, under which firms can choose the level of detail and the specific content of their disclosures. Relative to a rules-based approach, the regulator expects a principles-based approach to “minimise the risk of businesses publishing sanitised ‘generic’ tax strategies” (HMRC 2015c,

11). This expectation is consistent with recent research showing that firms' disclosure choices and characteristics can be informative about their tax planning. For example, Inger et al. (2018) find that firms engaging in greater (less) tax avoidance than industry peers have less readable (more straightforward) tax footnotes. Using firms providing sustainability disclosures, Adams et al. (2022) find a negative relation between the level of tax avoidance and the likelihood and amount of voluntary tax disclosures. Because a principles-based disclosure framework makes it possible for firms to reveal their "type" through disclosure choices, this disclosure approach should help shed light on firms' tax practices.

The two disclosure examples in Appendix A illustrate how a principles-based disclosure framework could be informative about a firm's tax practice and assist public scrutiny of tax avoidance. The first example is the initial tax strategy disclosure by Travis Perkins PLC—a FTSE 250 firm. The disclosure is detailed and contains abundant firm-specific information. For example, when discussing its governance structure and risk management protocols in relation to tax, this firm not only provides an overview of the procedures in place but also identifies the specific parties involved and their respective roles. The disclosure also highlights how the firm's reputation as a "responsible corporate citizen" influences its tax decisions. In comparison, the second disclosure by Marshalls PLC, also a FTSE 250 firm, is much shorter and less firm-specific. These examples demonstrate that disclosure characteristics such as the amount of information disclosed and the extent of firm-specific information could reflect a firm's effort to maintain a positive public image related to its tax practice. Detailed disclosures about the governance mechanisms in place could assure the public that the firm's actual tax practice aligns with the disclosure (GRI 2019).

Another important observation of Travis Perkins PLC's disclosure is the explicit statement that the firm has "a low risk appetite in relation to tax matters." In contrast, Marshalls PLC does

not provide such disclosure. Because it is costly for a firm to publicly acknowledge that it tolerates a high level of tax risk, a separating equilibrium is likely to occur where only firms that do not engage in any risky tax planning will explicitly disclose that they have a low tax risk appetite (Spence 1973; Verrecchia 1983). Thus, the existence of (or lack thereof) a tax risk appetite disclosure can be informative about a firm's tax practice.⁵ These two examples illustrate how certain disclosure choices and characteristics can be useful in gauging firms' willingness to engage in greater (or even more aggressive) tax avoidance and in scrutinizing those that likely do so.

The U.K. regulation requires firms to *publicly* disclose their tax strategy rather than privately communicate such information to the tax authority because the regulator believes that “reputational concerns can influence the attitudes of large businesses and encourage them to pursue less aggressive tax planning arrangements” (HMRC 2015b, 12). For example, the HMRC's qualitative interviews with key decision-makers show that public perception is an important consideration for tax-related decision-making in large businesses and that the potential impact of public perception is stronger for firms with a British identity (HMRC 2015a). As one interviewee stated, “This sense of [British] identity was used in boardroom negotiations about tax—the desire to retain a British identity and presence had a direct influence on tax strategy” (HMRC 2015b, 22). Anecdotal evidence also suggests that the U.K. has a strong and negative public sentiment toward corporate tax avoidance, which should work in favor of the regulation achieving its objective of curbing tax avoidance. For instance, after the revelation of Starbucks' tax avoidance schemes in 2012, a high-profile U.K. consumer boycott led to an unprecedented, voluntary tax payment of £20 million by the firm (Houlder and Thompson 2012; Neville and Treanor 2012). Similarly, a 2017 survey finds that 90 percent of U.K. adults consider corporate tax avoidance unethical despite

⁵ In fact, some firms even voluntarily disclose that they receive a low tax risk rating assessment from the HMRC. This evidence is consistent with firms using disclosures to signal their tax practices.

being legal and that over 40 percent would consider boycotting products and services from firms that do not pay their fair share of taxes (Pegg 2017). Thus, the regulator perceives public scrutiny as a powerful force that could assist the regulation in curbing tax avoidance (HMRC 2015a).

Effectiveness of the U.K. Regulation: Potential Concerns

Although the regulator expects that tax strategy disclosures will inform the public about firms' tax practices, thereby pressuring tax avoiders to change their behavior, several factors cast doubt on the effectiveness of this regulation in achieving its objective. First, while a principles-based disclosure approach could provide decision-useful information and reveal firm "type", this approach requires costly enforcement and can be counter-productive if firms take advantage of disclosure discretion (Leuz 2010). From the enforcement perspective, the regulation imposes financial penalties on non-compliance, and the dollar amount is much higher compared to other tax disclosure regulations in the U.K.⁶ However, these penalties may not be sufficiently punitive given the size of the firms subject to this regulation (i.e., at least £2 billion in total assets or £200 million in turnover). Because the regulation only requires that firms disclose some information about each mandatory disclosure component, the compliance threshold is low. If most firms respond to the disclosure by providing minimal information or using extensive boilerplate language, the disclosure will not be useful for identifying tax avoiders, which is a necessary condition for public scrutiny to successfully pressure firms to change tax practices.

Second, even in a country like the U.K., where there is a strong and negative public sentiment toward tax avoidance, it is still unclear whether and to what extent negative public perception can change firm behavior. In fact, research about the reputational costs of tax avoidance using non-U.K. settings provides mixed evidence. For example, Hanlon and Slemrod (2009)

⁶ For example, the maximum penalty for non-compliance with the U.K. subsidiary disclosure requirement examined in Dyreng et al. (2016) is only £1,000.

examine firms engaging in tax shelter transactions and find negative investor responses around the revelation of these egregious tax avoidance activities. However, Gallemore et al. (2014) use a similar setting and find that the negative market reactions documented in Hanlon and Slemrod (2009) reversed within weeks after the initial revelation, suggesting that the negative valuation impact of engaging in these transactions is only temporary. In addition, they do not find that tax shelter activities are associated with outcomes reflecting reputational costs such as changes in tax avoidance, auditor switches, manager turnover, and sales decrease. Thus, Gallemore et al. (2014) find little empirical support for tax reputational costs.

Exploiting different settings to capture the reputational costs of tax avoidance, Chen et al. (2019) do not find that large U.S. public firms change tax avoidance in response to negative media coverage. Hasegawa et al. (2013) examine publicly traded corporations in Japan and find no change in tax avoidance after these firms' tax return information is publicly disclosed. Hoopes et al. (2018) examine a similar regulation in Australia and find minimal changes in tax avoidance, even though equity investors anticipated net costs for firms that are likely to face increased public scrutiny (Chen 2017). In contrast, Kays (2022) finds that firms affected by the Australian regulation strategically increase voluntary disclosure to offset potential reputational costs, which may explain the lack of changes in tax avoidance.⁷ Overall, the mixed evidence casts doubt on the effectiveness of public scrutiny in reducing tax avoidance. Thus, it is unclear whether a novel public tax disclosure regulation could leverage public scrutiny to change firms' tax practices.

Lastly, the U.K. regulation did not create a common repository for the tax strategy

⁷ The Japanese and Australian tax return disclosure regulations examined in prior research focus on *quantitative* information, whereas the U.K. regulation mandates *qualitative* disclosures. In both the Japanese and the Australian settings, it is the tax authority that publishes tax return information. Firms subject to these regulations do not have control over the disclosure. In contrast, firms subject to the U.K. regulation not only disclose tax strategy themselves but also have the flexibility in choosing what to disclose and how much detail to disclose. Thus, the U.K. tax strategy disclosure requirement is fundamentally different from the public tax return disclosure regimes in Japan and Australia, suggesting that results from Hasegawa et al. (2013) and Hoopes et al. (2018) may not generalize to the U.K. setting.

disclosures. Allowing firms to publish tax strategy on the platform of their choice significantly increases the information awareness and acquisition costs for stakeholders who are interested in the disclosures, thereby weakening the level of actual public scrutiny these disclosures can attract (Blankespoor et al. 2019; Maines and McDaniel 2000). Taken together, the above arguments suggest that it is an empirical question whether the U.K. tax strategy disclosure requirement will achieve its objective of curbing tax avoidance. Thus, I state my hypothesis in the null form:

Hypothesis: The U.K. tax strategy disclosure requirement is not associated with changes in firms' tax avoidance.

III. RESEARCH DESIGN AND SAMPLE SELECTION

Difference-in-differences (DiD) Design

I use a difference-in-differences (DiD) research design to examine the primary research question. Specifically, I estimate the following OLS model:

$$\begin{aligned} Tax_Avoidance = & \alpha_0 + \beta_1 Post \times Treat + \beta_2 Size + \beta_3 Leverage + \beta_4 Intangible + \\ & \beta_5 Inventory + \beta_6 R\&D + \beta_7 PPE + \beta_8 Capex + \beta_9 ROA + \\ & \beta_{10} Loss + Firm\ FE + Year\ FE + \epsilon \end{aligned} \quad (1)$$

The dependent variable (*Tax_Avoidance*) is either one-year 1) book effective tax rate (*ETR*) or 2) cash effective tax rate (*CETR*). I define *ETR* (*CETR*) as income tax expense (cash taxes paid) scaled by pretax income before special items (Dyreng, Hanlon, and Maydew 2008). Both measures decrease in tax avoidance. Book ETR reflects tax avoidance for financial reporting purposes and is relevant to my setting due to its saliency to the public (Graham et al. 2014; Austin and Wilson 2017; Chen et al. 2019; Rego et al. 2020). In comparison, cash ETR reflects cash tax payments and better captures the HMRC's goal to increase tax revenue than book ETR does.⁸

⁸ Prior research sometimes finds asymmetric results when using book versus cash ETRs as the measure of tax avoidance. For example, Robinson, Sikes, and Weaver (2010) find that firms evaluating the tax department as a profit center are associated with greater tax avoidance than those evaluating the tax department as a cost center. However, this finding only manifests when using book ETR as the measure of tax avoidance, which the authors attribute to

Post is an indicator variable equal to one for fiscal years beginning after September 15, 2016, when the U.K. tax strategy disclosure requirement became effective. To identify firms that are subject to this regulation, I compare firms' total assets and turnover in the year immediately prior to the regulation becoming effective to the disclosure thresholds.^{9,10} I set an indicator variable, *Treat*, equal to one if a firm meets at least one disclosure threshold (i.e., treated firm), and zero otherwise (i.e., control firm).¹¹ The interaction term, $Post \times Treat$, is the variable of interest and captures the differential changes in tax avoidance for the treated firms in the post-regulation period relative to the control firms. If the regulation is effective in curbing tax avoidance, I expect a positive coefficient on the interaction term ($\beta_l > 0$). Following prior research (e.g., Gupta and Newberry 1997; Rego 2003; Chen et al. 2010), I control for the determinants of tax avoidance, including size (*Size*), leverage (*Leverage*), intangibles (*Intangible*), inventory (*Inventory*), research and development activities (*R&D*), tangible assets (*PPE*), capital expenditures (*Capex*),

accrual tax avoidance. Because tax planning takes time to implement, it is possible that firms can adjust book ETRs (through tax accruals) faster than cash ETRs in response to the U.K. regulation. Thus, I examine both book and cash ETRs to account for potential asymmetric responses.

⁹ For firms with fiscal year-end months in 1/2/3/4/5/9/10/11/12, the regulation became effective in 2017 fiscal year. For these firms, *Post* includes fiscal years 2017 through 2019, and *Treat* is constructed using fiscal year 2016 financial statements. For firms with fiscal year-end months in 6/7/8, the regulation became effective in 2018 fiscal year. Thus, *Post* includes fiscal years 2018 and 2019, and *Treat* is constructed using fiscal year 2017 financial statements.

¹⁰ As discussed in "Sample Section", my sample includes U.K.-incorporated firms. For purposes of this U.K. regulation, the U.K. parent company and its subsidiaries form a "U.K. group". To determine whether a U.K. group is subject to the disclosure requirement, Schedule 19, Part 1, Sections 10(2)-10(3) of Finance Act 2016 requires the group to use "group" assets and turnover, defined as aggregate total assets and turnover across all U.K. companies within the group (Finance Act 2016, 590). "U.K. companies" further include U.K. subsidiaries and U.K. permanent establishments of the non-U.K. subsidiaries in the group. Thus, the language of the regulation suggests that the disclosure thresholds are based on the "U.K. companies" in the "U.K. group". However, nearly 70 percent of the sample firms only file consolidated financial statements that include both U.K. and non-U.K. entities. Therefore, I use consolidated total assets and turnover to identify treated firms, acknowledging that these measures are imperfect. In a concurrent working paper, Bilicka et al. (2022) also use worldwide consolidated financials to identify treated firms for the same reason. In an untabulated test, I find similar results if I exclude treated firms that did not disclose tax strategy. Because these firms are most susceptible to potential misclassification by using consolidated financials, this test should alleviate the concern that imperfect classification of treated firms might explain the primary findings.

¹¹ In untabulated analysis, I confirm that about 94 percent of the firms subject to the disclosure requirement in the first year when it became effective continue to meet at least one disclosure threshold over the remaining years in post-regulation period. Thus, *Treat* is "sticky" over time.

profitability (*ROA*), and prior losses (*Loss*). I include firm fixed effects to control for time-invariant, unobservable firm characteristics, and I include year fixed effects to control for time-varying macroeconomic factors. Given the fixed effects structure, the main effects of *Post* and *Treat* are subsumed. Appendix B provides detailed variable definitions.

Cross-sectional Tests within Treated Firms

I complement the DiD design with a series of cross-sectional tests *within* treated firms (Chen et al. 2019; De Simone et al. 2022). These tests offer two empirical benefits: First, the cross-sectional tests focus on subsamples that are most likely to exhibit the intended behavioral changes, whereas the DiD design estimates the average effect of the regulation across *all* treated firms. Thus, the cross-sectional tests allow me to capture the effect of the regulation, even if it only exists in subsamples of treated firms. Second, within-treatment analysis alleviates the concern that the results of the DiD analysis could be driven by changes in tax avoidance among control firms. I modify equation (1) by replacing the indicator variable for treated firms (*Treat*) with a set of cross-sectional variables (*XS_VAR*) to capture subsamples that are most likely to reduce tax avoidance in response to the U.K. regulation. Specifically, I estimate the following OLS model:

$$\begin{aligned} Tax_Avoidance = & \alpha_0 + \beta_1 Post \times XS_Var + \beta_2 Size + \beta_3 Leverage + \beta_4 Intangible + \\ & \beta_5 Inventory + \beta_6 R\&D + \beta_7 PPE + \beta_8 Capex + \beta_9 ROA + \\ & \beta_{10} Loss + Firm\ FE + Year\ FE + \epsilon \end{aligned} \quad (2)$$

I first partition treated firms based on their ownership of (or lack thereof) valuable brands. *Brand* is an indicator variable set equal to one if the firm owns at least one valuable brand based on Kantar Millward Brown's Top 50 (Top 75) U.K. brands in 2017 (2018) (Kantar 2017, 2018), and zero otherwise. I consider valuable brands relevant because firms with valuable brands are more visible to the public. Accordingly, their tax strategy disclosures are more likely to attract public scrutiny than firms without (Austin and Wilson 2017; Chen et al. 2019). Prior research

suggests that consumers have a more unanimous, negative view toward tax avoidance than other stakeholders, and firms with valuable brands are more sensitive to negative public perceptions of their tax practices than those without due to the impact on brand value (Austin and Wilson 2017). Thus, all else equal, I expect that firms with valuable brands are more likely to change tax behavior in response to the public scrutiny of their tax strategy disclosures than those without ($\beta_1 > 0$).

Next, I partition treated firms based on the presence of tax haven operations. *Haven_Sub* is an indicator variable set equal to one if the firm has at least one subsidiary in a tax haven country based on Bureau van Dijk's Orbis database, and zero otherwise. I define tax haven countries following Dyreng and Lindsey (2009). Dyreng et al. (2016) show that the U.K. public negatively perceives tax haven operations, and firms decrease their use of tax havens after being subject to negative publicity surrounding their subsidiary disclosures. Thus, firms with tax haven operations likely face greater public scrutiny than those without. Accordingly, I expect the regulation to have a greater impact, if any, on firms with tax haven operations than those without ($\beta_1 > 0$).

Sample Selection

I use the Compustat Global database to construct a sample of U.K.-incorporated firms listed on the London Stock Exchange from 2010 to 2019. Because the U.K. enacted a major tax reform and switched to a territorial corporate tax system in 2009, I set the sample period to begin in 2010 to hold the tax regime constant. Focusing on U.K.-incorporated firms offers several empirical benefits. First, U.K. firms are more likely and willing to change tax behavior than non-U.K. firms due to their British identity (HMRC 2015b). Thus, using a sample of U.K.-incorporated firms should increase the power of detecting changes in tax avoidance, if any. Second, the level of public scrutiny and potential enforcement of this regulation likely differ between U.K. and non-U.K. firms. Focusing on U.K.-incorporated firms holds the tax environment and country institutions

constant across sample firms. Finally, because the dependent variables are effective tax rates, this sample selection choice alleviates a potential concern that the results are driven by variations in the statutory tax rates across countries. I exclude mutual funds because they are taxed differently. I also remove member firms of a group because each qualifying group only provides one tax strategy disclosure, making it difficult to attribute any behavioral change at the group level to individual member firms. I further require non-missing variables to estimate equation (1). Lastly, I retain firms that appear in both the pre- and post-regulation periods. The final sample using book (cash) ETR as the measure of tax avoidance includes 6,920 (5,080) observations related to 864 (624) unique firms. Table 1 provides detailed sample selection procedures.

IV. RESULTS

Difference-in-differences Design

Table 2, Panel A presents descriptive statistics for the variables used to estimate equation (1). Treated (control) observations constitute 52.0 (48.0) percent of the regression sample. On average, sample firms report *ETR* (*CETR*) of 19.2 (21.4) percent and are profitable with an 8.2 percent return on assets (*ROA*). Panel B presents descriptive statistics separately for the treated ($Treat = 1$) and control ($Treat = 0$) subsamples. As expected, treated firms are larger (*Size*) than control firms at the mean (i.e., 7.85 versus 4.35). Treated firms also differ from control firms in other firm characteristics such as leverage (*Leverage*), tangibility (*PPE*), etc.¹²

Table 3, Panel A presents the results of estimating equation (1) using the DiD design. For completeness, I report regression results both with and without control variables. In columns (1) and (2), where the dependent variable is *ETR*, the coefficient estimates on $Post \times Treat$ are

¹² These univariate statistics could suggest that control firms may not be ideal counterfactuals to treated firms due to differences in firm characteristics, especially in terms of firm size. In a supplemental test, I address this concern using a regression discontinuity design to provide corroborating evidence.

statistically insignificant. I confirm that the 95 percent confidence intervals for the estimated coefficients on $Post \times Treat$ include zero in both columns, which provides evidence that is compatible with treated firms not exhibiting differential changes in tax avoidance post-regulation relative to control firms (Cready et al. 2022). The upper bound (UB) estimate is particularly relevant to this study because it is in the predicted direction if the regulation is effective. Using the estimate from column (2) as an example, the UB of the 95 percent confidence interval for $Post \times Treat$ is 0.023, or 11.88 (13.55) percent of the mean (standard deviation) of the dependent variable (Cready et al. 2022). To reach statistical significance at the ten percent level using two-tailed tests, the coefficient estimate on $Post \times Treat$ has to be 0.019, or 375 percent larger than the current estimate of 0.004. Collectively, these statistics are most compatible with treated firms not differentially reducing tax avoidance after the regulation became effective.

Inferences are similar when $CETR$ is the dependent variable. In columns (3) and (4), the coefficient estimates on $Post \times Treat$ are not only statistically insignificant but also opposite to the predicted direction if the regulation is effective in curbing tax avoidance. In both columns, the 95 percent confidence intervals include zero. Using the estimate from column (4) as an example, the UB of the 95 percent confidence interval for $Post \times Treat$ is 0.012, or 5.39 (6.35) percent of the mean (standard deviation) of the dependent variable. To reach statistical significance at the ten percent level using two-tailed tests, the coefficient estimate on $Post \times Treat$ has to be 0.025, or 278.57 percent larger than the current estimate. Although the statistically insignificant estimates are not sufficient evidence to conclude that the regulation is ineffective in changing firms' tax avoidance, findings are consistent across all specifications and are most compatible with the

regulation not having a meaningful impact on treated firms' tax practices.¹³

An underlying assumption of the DiD design is that tax avoidance for treated and control firms would evolve in parallel absent the treatment effect (i.e., parallel trend assumption). Although this assumption cannot be empirically confirmed, I conduct two tests to evaluate the reasonableness of this assumption in my setting following Armstrong et al. (2022). Figure 1 separately plots the time-series trend of tax avoidance for treated and control firms using *ETR* (Panel A) and *CETR* (Panel B). In both panels, I do not observe significant differences or different trends in tax avoidance between treated and control firms in the pre-regulation period. Next, I re-estimate equation (1) after replacing *Post* with separate indicators for individual years in the sample period. I use the year immediately prior to the regulation becoming effective (i.e., $t-1$) as the benchmark. Table 3, Panel B presents results. For parsimony, I tabulate results for the specification that includes the control variables. Inferences are similar for the specification without control variables. In both columns, the coefficient estimates on the interaction terms are mostly statistically insignificant in years prior to the regulation becoming effective. Thus, I do not observe significant pre-treatment differences in tax avoidance between treated and control firms. Taken together, the evidence suggests that the parallel trend assumption is plausible in my setting.

Within-treatment, Cross-sectional Tests

Valuable Brands

Table 4, Panel A presents the results of estimating equation (2) after partitioning treated firms based on ownership of (or lack thereof) valuable brands (*Brand*). In columns (1) and (2),

¹³ Unlike this study and Bilicka et al. (2022), both of which do not find evidence indicative of changes in tax avoidance after the U.K. tax strategy disclosure regulation became effective, a subsequent working paper by Dunker and Willkomm (2022) provides mixed evidence. However, the authors disclose that if they follow a similar empirical approach used in this study and in Bilicka et al. (2022), they “also do not find an effect of the regulation on tax avoidance” (Dunker and Willkomm 2022, 5).

where *ETR* is the measure of tax avoidance, the coefficient estimates on *Post*×*Brand* are statistically insignificant and opposite to the predicted effect of the regulation. In both columns, the corresponding 95 percent confidence intervals include zero. Using the estimate from column (2) as an example, the UB of the 95 percent confidence interval for *Post*×*Brand* is 0.024, or 11.61 (15.29) percent of the mean (standard deviation) of the dependent variable. Inferences are similar in columns (3) and (4), where *CETR* is the measure of tax avoidance. These findings are incompatible with differential changes in tax avoidance in the post-regulation period between firms with valuable brands and those without.

Austin and Wilson (2017) show that the reputational costs of tax avoidance vary by branding strategy. Specifically, firms with a “house of brands” strategy (i.e., brand names are unrelated to the name of the owner firm) are less sensitive to reputational concerns than those with a “corporate branding” (i.e., brand name is similar to the name of the owner firm) or a mixed branding strategy. This is because a “house of brands” strategy creates a weaker connection between the valuable brands and the owner firm relative to the other two strategies. Motivated by this finding, I further decompose *Brand* into *Brand_House* and *Brand_Other* to explore potential variation based on branding strategy. I set *Brand_House* (*Brand_Other*) equal to one if the firm adopts a “house of brands” strategy (a “corporate branding” or a mixed branding strategy), and zero otherwise.¹⁴ Following Austin and Wilson (2017), I expect that firms with a corporate branding or a mixed branding strategy (*Brand_Other* = 1) are more likely to exhibit the intended behavioral changes, if any, than firms with a “house of brands” strategy (*Brand_House* = 1).

Panel B of Table 4 presents the results. Across all specifications, the coefficient estimates

¹⁴ Among sample firms with a valuable brand, Burberry Group PLC adopts a “corporate branding” strategy and its valuable brand is “Burberry”. Unilever PLC uses a “house of brands” strategy and owns valuable brands “Dove” and “Lipton”. Royal Bank of Scotland uses a mixed branding strategy with two valuable brands: “RBS” and “NatWest”.

on $Post \times Brand_Other$ are statistically insignificant. In three out of the four columns, the coefficient estimates are opposite to the predicted direction if the regulation is effective. In all four columns, the coefficient estimates on $Post \times Brand_Other$ are not statistically different from those on $Post \times Brand_House$, which is inconsistent with firms using a “corporate branding” or a mixed branding strategy being more likely to attract public scrutiny and change behavior. Taken together, Table 4 results are most compatible with firms having a valuable brand not differentially reducing tax avoidance in the post-regulation period, even though these firms should be the most sensitive to public scrutiny of their tax practices.

Tax Haven Operations

Table 5 presents the results of estimating equation (2) after partitioning treated firms based on the presence of tax haven operations ($Haven_Sub$). In all four columns, the coefficient estimates on $Post \times Haven_Sub$ are statistically insignificant and opposite to the predicted direction if the regulation is effective. In addition, the corresponding 95 percent confidence intervals include zero. Using the estimate from column (2) as an example, the UB of the 95 percent confidence interval for $Post \times Haven_Sub$ is 0.018, or 9.02 (13.08) percent of the mean (standard deviation) of the dependent variable. Inferences are similar in columns (3) and (4), where $CETR$ is the measure of tax avoidance. Thus, Table 5 results are most compatible with firms having tax haven operations not differentially reducing tax avoidance post-regulation, even though these firms are more likely to attract public scrutiny of their tax strategy disclosures. Collectively, findings in Tables 4 and 5 are inconsistent with the regulation having achieved its objective.

V. SUPPLEMENTAL AND ROBUSTNESS TESTS

Regression Discontinuity Design

Because the U.K. tax strategy disclosure requirement uses total assets and turnover as

disclosure thresholds, a primary concern of the DiD research design is that control firms may not be comparable to treated firms. To corroborate the primary results, I examine changes in tax avoidance in response to this regulation using a sharp regression discontinuity (RD) design as a supplemental test. An RD design uses observations that are within a certain “bandwidth” around the disclosure threshold to estimate the treatment effect. The empirical appeal of implementing this design in the U.K. setting is that, relative to firms that are further away from the disclosure threshold, firms just below the disclosure threshold (and therefore untreated) provide a better counterfactual to those immediately above the threshold (and treated). Thus, an RD design directly addresses the primary concern of the DiD approach.¹⁵

To implement the RD design, I restrict the primary sample to the post-regulation period because the treatment effect does not exist prior to the regulation becoming effective. I use *RV_Turnover* as the “running” variable, to which I apply the treatment rule to assign each observation to the treatment and control group based on a threshold cutoff (Cattaneo, Idrobo, and Titiunik 2019). Specifically, *RV_Turnover* equals the difference between the firm’s turnover, measured in the year immediately prior to the regulation taking effect, and the £200 million turnover disclosure threshold (De Simone and Olbert 2022).¹⁶ An important assumption of the RD design is that individual observations are unable to precisely manipulate the assignment variable (i.e., turnover) to receive or avoid the treatment effect (Cattaneo et al. 2019; Lee and Lemieux 2010). In an untabulated analysis, I plot the distribution of the running variable and find that the

¹⁵ Entropy balancing is not feasible in this setting because it is impossible to achieve covariate balance between treated and control firms if *Size* is included as a covariate. For a similar reason, a propensity score matching (PSM) approach is not implementable if *Size* is a matching criterion. Although I can implement either entropy balancing or PSM without including *Size*, doing so does not address the size difference between treated and control firms.

¹⁶ Although either total assets or turnover could determine whether a firm is subject to the U.K. regulation, the turnover threshold is a more common trigger of the treatment effect than the total assets threshold. Specifically, only 68 percent of the treated firms meet both thresholds. For the remaining 32 percent, most firms (i.e., 30 percent) only meet the turnover threshold (and not the total assets threshold). Thus, turnover is more important in determining whether a firm receives the treatment effect. Accordingly, I use turnover to construct the running variable.

95 percent confidence intervals for the treated and control firms on both sides of the threshold overlap (Cattaneo et al. 2019), a pattern that is consistent with no manipulation of the disclosure threshold. In addition, I formally test the density discontinuity for *RV_Turnover* and fail to reject the null hypothesis of continuity of the density function at the threshold. Thus, there is no statistical evidence suggesting manipulation of turnover at the disclosure threshold. These findings also help alleviate the potential concern that the primary findings could be attributable to firms manipulating disclosure thresholds to avoid being subject to the disclosure requirement.

Following the recommendation of existing research (Cattaneo et al. 2019; De Simone and Olbert 2022; Lee and Lemieux 2010), I estimate both local linear and second-order polynomial regressions with a triangular kernel function that assigns greater weights to observations in closer proximity to the disclosure threshold. I select optimal bandwidths that minimize the mean-squared error of the local RD estimator (Cattaneo et al. 2019). Table 6 presents the results. In all four columns, the RD estimates are statistically insignificant, which is most compatible with treated firms not differentially reducing tax avoidance in the post-regulation period. More importantly, the findings of the RD analysis corroborate the results of the DiD analysis.

U.K. Tax Avoidance

In the primary analyses, I measure tax avoidance using *ETR* and *CETR*, both of which capture firms' worldwide tax avoidance. Because the regulation mandates information in relation to U.K. taxation, the primary objective of the regulation is to curb tax avoidance that reduces U.K. tax revenues (i.e., U.K. tax avoidance). If treated firms respond to potential increases in public scrutiny of *U.K. tax avoidance* by shifting such activity to other jurisdictions, they could increase U.K. tax expense or payments without affecting their worldwide tax burden (De Simone, Stomberg, and Williams 2023). In this supplemental test, I examine whether this scenario explains

the primary findings by re-estimating equation (1) within a subsample of U.K. domestic firms, for which *ETR* and *CETR* reflect U.K. tax avoidance. I identify U.K. domestic firms as those that do not have any non-U.K. subsidiaries based on Bureau van Dijk's Orbis database. In Table 7, across all specifications, the coefficient estimates on $Post \times Treat$ are statistically insignificant, and the corresponding 95 percent confidence intervals include zero. Thus, the primary inferences continue to hold, and findings are most compatible with no differential changes in U.K. tax avoidance between treated and control firms following the regulation.

Country-by-Country (CbC) Reporting

The U.K. adopted CbC reporting for fiscal years starting on or after January 1, 2016 (HMRC 2019). Firms with global turnover exceeding €750 million and operations in both the U.K. and at least one other country are subject to this reporting requirement (HMRC 2019). The first CbC report is due in 2017, which coincides with the year when the U.K. tax strategy disclosure requirement became effective. The regulator views the CbC regime as “an additional risk-assessing tool for tax administrations” (HMRC 2015b, 12), whereas it considers the U.K. tax strategy disclosure requirement “separate to, and distinct from, the OECD's country-by-country reporting model” (HMRC 2015b, 13). Thus, these two disclosure regimes are inherently different. Nevertheless, to empirically rule out any potential confounding effect of the CbC regime, I exclude firms that are subject to mandatory CbC reporting from the sample and re-estimate equation (1). In Table 8, across all specifications, the coefficient estimates on $Post \times Treat$ remain statistically insignificant, and the corresponding 95 percent confidence intervals include zero. These findings suggest it is unlikely that the CbC reporting requirement explains the primary results.

Robustness Tests

I conduct several robustness tests (untabulated). First, some FTSE 50 firms had voluntarily

disclosed information about tax strategy prior to the U.K. regulation becoming effective (HMRC 2015a). Because a rational firm would only provide such disclosure voluntarily if the disclosure costs are low, I do not expect tax avoiders—the target of the U.K. regulation—to do so prior to the disclosure mandate. Thus, if the regulation is effective, it should still be able to attract public scrutiny of tax avoiders and pressure for behavioral changes. For robustness, I exclude treated firms that voluntarily disclosed information about tax strategy in the pre-regulation period and re-estimate equation (1).¹⁷ Inferences are similar, suggesting that rare cases of voluntary disclosure in the pre-regulation period do not explain the lack of changes in tax avoidance post-regulation. Second, I re-estimate equation (1) with long-run (i.e., three-year) book and cash ETRs after excluding fiscal years for which the ETR measurement window spans both the pre- and post-regulation periods, and I find similar inferences.¹⁸ These robustness tests suggest that the primary results are not attributable to the empirical proxies used or certain research design choices.

VI. EXPLORATORY ANALYSIS

Exploratory Analysis

The evidence thus far is largely inconsistent with the regulator’s expectation that the disclosure requirement will curb tax avoidance. In this section, I examine select factors that could influence the effectiveness of the regulation. I consider the analysis exploratory and acknowledge that the evidence is largely indirect. However, such evidence could offer potential explanations for the lack of empirical findings that firms reduce tax avoidance in response to the U.K. regulation.

¹⁷ I identify firms that voluntarily disclosed tax strategy in the pre-regulation period using hand-collected data. I consider disclosures labeled as “Tax Policies”, “Principles on Tax”, etc. to be related to tax strategy. Because this process involves judgment, I also repeat the analysis after excluding all FTSE 50 firms and find similar results.

¹⁸ In an untabulated analysis, I examine tax risk, measured as three-year book or cash ETR volatility, as an alternative outcome variable (Guenther, Matsunaga, and Williams 2017). To prevent the three-year measurement window of the tax risk proxies from spanning both the pre- and post-regulation periods, which could introduce noise in the measures, I only retain year 2019 in the post-regulation period. Similarly, I restrict the pre-period to year 2016 to offer a balanced pre- and post-window. I re-estimate equation (1) using this subsample but do not find statistically significant evidence suggesting differential changes in tax risk for treated firms relative to control firms in the post-regulation period.

Information Acquisition Costs

I first consider information acquisition costs. To replicate the experience of an average stakeholder (such as a consumer) who is interested in accessing firms' tax strategy disclosures, I hand-collect treated firms' disclosures from corporate websites. To allow for tractability in data collection, I focus on tax strategy disclosures in the first year when they are mandated. For each treated firm, I start by searching the term "tax strategy" on the corporate website. If the initial search is unsuccessful, I complement it with Google searches using the company name plus the term "tax strategy" to locate the disclosure.¹⁹ Out of the 458 treated firms in the main regression sample, I am able to obtain initial tax strategy disclosures for only 204 firms.^{20,21} Nearly all firms publish their tax strategy on the corporate website, but these disclosures are published in various sections including the "sustainability" and the "investor relations" page. The complex search process to locate a firm's tax strategy disclosure echoes the concern that the lack of a common repository for these disclosures likely imposes nontrivial information awareness and acquisition costs on stakeholders who are interested in accessing these disclosures (Maines and McDaniel 2000; Blankespoor et al. 2019).

Disclosure Choices and Characteristics

The regulator believes that a principles-based disclosure approach should maximize

¹⁹ Where necessary, I use Wayback Machine (<https://web.archive.org/>) to retrieve previously published tax strategy disclosures that were no longer available on the corporate website at the time of data collection.

²⁰ When constructing the primary regression sample, I do not require a treated firm to be compliant with the U.K. regulation (i.e., disclosed tax strategy) because a firm's choice to comply (versus not) could be informative of its tax practice. Specifically, if tax avoiders are more likely to be non-compliant with the disclosure requirement due to higher disclosure costs than non-tax avoiders, I expect that non-compliant firms will attract greater public scrutiny and are more likely to change behavior. In an untabulated analysis, I partition treated firms based on their compliance status. Across all specifications examined, I continue to find results that are most compatible with no differential changes in tax avoidance conditional on firms' compliance status.

²¹ In an untabulated analysis, I follow the same procedures to examine whether control firms might voluntarily provide U.K. tax strategy disclosures. Out of the 456 control firms in the primary sample, only 25 firms (or 5.48 percent of the control subsample) disclosed tax strategy in the post-regulation period. Inferences are similar if I exclude these control firms from the regression sample.

information useful to the public and that qualitative tax strategy disclosures can be informative about firms' tax practices. Motivated by these expectations, I next explore the information content and characteristics of the 204 hand-collected initial tax strategy disclosures. I find that firms disclose 3.37 (out of four) required components on average, suggesting a high level of compliance conditional on firms disclosing their tax strategy. Nearly all firms disclose information about tax risk management (99 percent), attitude toward tax planning (96 percent), and the approach to dealing with the HMRC (93 percent). At the mean, firms' disclosures contain about 899 words, but the length of the disclosures ranges from 294 to 3,862 words (untabulated), suggesting significant variations in the amount of information disclosed.

Interestingly, only 49 percent of the firms explicitly disclosed tax risk appetite. The lower compliance rate suggests that firms perceive this mandatory disclosure component as more costly than the other three. It is also consistent with a potential separating equilibrium where only firms with a low tax risk appetite will publicly disclose this component. Thus, the absence of the tax risk appetite disclosure could indicate that the firm tolerates more or even riskier tax avoidance under a mandatory disclosure regime. I re-estimate equation (2) after partitioning treated firms based on whether they explicitly disclosed tax risk appetite. If the public understands the implications of firms' disclosure choices related to tax risk appetite, I expect firms not disclosing this component (*NoDisc_RiskAppt*) to attract greater public scrutiny and thus, are more likely to change behavior than firms disclosing their tax risk appetite. In Table 9, the coefficient estimates on the interaction terms are statistically insignificant and opposite to the predicted effect of the regulation in all columns. I confirm that the corresponding 95 percent confidence intervals include zero. Using the estimate from column (2) as an example, the UB of the 95 percent confidence interval for $Post \times NoDisc_RiskAppt$ is 0.012, or 8.58 (6.04) percent of the mean (standard deviation) of the

dependent variable. Taken together, these statistics are most compatible with no differential changes in tax avoidance conditional on firms' disclosure choices related to tax risk appetite, which could reveal firms' "type".

Recent studies show that firms' tax avoidance is associated with disclosure characteristics (e.g., Inger et al. 2018; Adams et al. 2022). In untabulated analyses, I examine whether select disclosure characteristics could inform the public about firms' tax practices, which is a necessary condition for public scrutiny to pressure for behavioral changes. I re-estimate equation (2) after partitioning treated firms based on the amount of information disclosed (measured by the length of the disclosure) and the extent of firm-specific information disclosed (measured by the extent of boilerplate language). Because firms evaluate the costs and benefits when making disclosure choices (Belnap 2023), under a mandatory disclosure regime, I expect tax avoiders to provide shorter and more boilerplate disclosures than non-avoiders. If the public understands the implications of these disclosure characteristics, they should be able to identify firms that are likely to engage in greater tax avoidance. Again, results are most compatible with no differential changes in tax avoidance. Taken together, the findings of these analyses could be consistent with the possibility that the public does not fully understand the implications of firms' disclosure choices and characteristics for the underlying tax practices, which may explain the lack of changes in tax avoidance documented in the primary analysis. In a recent study, Blaufus, Jarzembski, Reineke, and Trenn (2023) examine how firms use tax strategy disclosures to portray themselves as either "tax planners" or "responsible citizens" but find little association between firms' narratives and tax avoidance measured by ETRs. They interpret the findings as tax strategy disclosures not being sufficiently informative about firms' underlying tax practices, which could offer another *ex-post* explanation for the lack of behavioral changes among firms subject to this U.K. regulation.

VII. CONCLUSION

In 2016, a new U.K. regulation requiring large businesses to publicly disclose their tax strategy received legislative approval. The regulator expects these mandatory disclosures to provide useful information to the public in scrutinizing firms' tax practices, thereby curbing corporate tax avoidance (HMRC 2015a). Motivated by a growing demand for greater tax transparency worldwide and a rising interest in using tax disclosures to constrain tax avoidance, this study examines whether the U.K. tax strategy disclosure requirement is effective in achieving its objective. Using a difference-in-differences design and a series of within-treatment, cross-sectional tests, my findings are most compatible with the regulation not having a significant impact on firms' tax avoidance. Thus, I provide direct evidence that is inconsistent with the regulator's belief that this U.K. regulation—at least in its current design—will successfully curb tax avoidance. To my knowledge, this study is among the first to examine a mandatory, qualitative public tax disclosure requirement. The findings of this study should inform regulators who are interested in implementing similar disclosure initiatives to combat tax avoidance.

This study is subject to three limitations. First, I only examine changes in tax avoidance, which is the stated objective, as the consequence of the regulation. Thus, this study alone cannot draw conclusions about the net benefits or costs of the regulation. Second, this study examines U.K. publicly traded corporations, and the findings of this study may not generalize to firms incorporated outside the U.K., private firms, or non-corporate entities affected by this regulation. Finally, estimating statistically insignificant changes in tax avoidance among firms affected by the regulation does not constitute definitive evidence that the intended behavioral changes do not exist. Instead, I can only conclude that the collective evidence in this study is most compatible with the regulation not being effective in curbing tax avoidance.

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Appendix A: Examples of U.K. Tax Strategy Disclosures

This appendix includes two illustrative examples of the U.K. tax strategy disclosures.

Example One

Below is the initial tax strategy disclosure by Travis Perkins PLC, a retailer of building and home improvement materials. Travis Perkins PLC is a FTSE 250 firm listed on the London Stock Exchange. The following disclosure can be accessed via the link below:

<https://www.travisperkinsplc.co.uk/~media/Files/T/Travis-Perkins/group-tax-strategy-november-2017.pdf>

Tax Strategy

Scope

The Group is headquartered in the UK. The vast majority of the Group's operations are UK based, with only four small overseas subsidiaries.

This strategy applies to Travis Perkins plc and to the group of companies headed by Travis Perkins plc in accordance with paragraph 16 of Schedule 19 to the Finance Act 2016. A list of the entities to which it applies is set out below. In this strategy, references to 'Travis Perkins plc' and 'the group' are to all these entities.

This strategy applies from the date of publication until it is superseded. References to 'UK Taxation' are to the taxes and duties set out in paragraph 15(1) of the Schedule which include Income Tax, Corporation Tax, PAYE, NIC, VAT, Insurance Premium Tax, and Stamp Duty Land Tax. References to 'tax', 'taxes' or 'taxation' are to UK taxation and to all corresponding worldwide taxes and similar duties in respect of which the Group has legal responsibilities.

Aim

The Group pays in its own right, and collects from third parties on behalf of the Government, substantial amounts of UK tax and these are remitted to the UK exchequer as they fall due.

The Group's objectives in managing and controlling its tax affairs and related tax risks are to:

- Ensure compliance with all applicable rules, regulations and disclosure requirements under which the Group operates;
- Develop and maintain an open and co-operative relationship with the tax authorities it deals with.
- Be transparent about its dealings with these tax authorities;
- Pay the correct amount of tax as it falls due;
- Maintain appropriate systems, processes and controls to ensure the correct calculation and recording of tax liabilities and receivables;
- Protect its reputation as a responsible UK tax payer; and
- Protect its ability to service government contracts when such opportunities arise.

Governance

The Board is responsible for the Group's tax strategy and policies and has delegated responsibility for their implementation within the Group to the Group's Chief Financial Officer (CFO). As part of the annual financial reporting process the CFO, who is also the Group's Senior Accounting Officer, reports to the Board of Directors on the Group's compliance with its tax strategy, policies and on its tax position.

Every six months the Audit Committee receives an update on the status of the Group's tax position, its effective tax rate, tax provisions and key tax issues to ensure it is fully informed of tax developments and issues affecting the Group.

Tax risk management

Tax risks are assessed as part of the Group's formal governance process and are reviewed by the CFO and reported to the Audit Committee and Board on a regular basis.

Significant tax risks, implications arising from these risks and potential mitigating actions are considered by the Group Board in line with the Group's governance process.

Appendix A: Examples of U.K. Tax Strategy Disclosures (continued)

Example One (continued)

The Group's governance structure requires that specialist third party technical advice is obtained on any significant tax treatments before the Board of Directors authorises the tax position to be adopted by the Group.

The systems used to determine tax liabilities are subject to regular review by the Internal Audit Department whilst the external auditors undertake a review of the Group's corporation tax provisions as part of the annual statutory audit. HMRC periodically reviews aspects of the Group's systems which the Group welcomes as it provides further assurance that tax is being paid as it falls due.

Approach to Tax Planning and Level of Risk

The Group seeks to protect its reputation as a responsible corporate citizen by ensuring that it acts in accordance with the letter and the spirit of current tax legislation so that it pays the right amount of tax when it falls due.

The level of risk which the Group accepts in relation to UK taxation is consistent with its overall objective of achieving certainty in the group's tax affairs. In relation to any specific issue or transaction, the Board is ultimately responsible for identifying the risks, including tax risks, which need to be addressed and for determining what actions should be taken to manage those risks, having regard to the materiality of the amounts and obligations in question. The Group considers that it has a low risk appetite in relation to tax matters.

Relationships with the Tax Authorities

The Group's policy is to deal with all tax authorities in an open, professional and mutually respectful manner. The Group employs professional advisors to assist it with complex areas of its tax affairs and to deal with the tax authorities in the overseas jurisdictions in which it operates.

Tax legislation is complicated and can be difficult to interpret. As a result it can take time to reach an agreed position with the tax authorities. The Group aims to work collaboratively with the tax authorities to achieve certainty of its tax position and to resolve open tax enquiries and disputes as rapidly as possible, whilst recognising its responsibility to protect the interests of its stakeholders.

The Group meets with HMRC annually to discuss the Group's risk rating and ensure it is being maintained at a suitable level.

In large businesses governed by complex tax legislation, adjustments are occasionally discovered by the Group to the amount of tax previously paid. In these circumstances the Group's policy is to inform HMRC and promptly pay any tax due or reclaim any tax refund.

Further information

The Group's approach to tax applies to all entities in the Group and to all transactions it enters into with third parties outside the Group. The Group's tax strategy is reviewed and updated annually as appropriate.

Approved by: Alan Williams, CFO

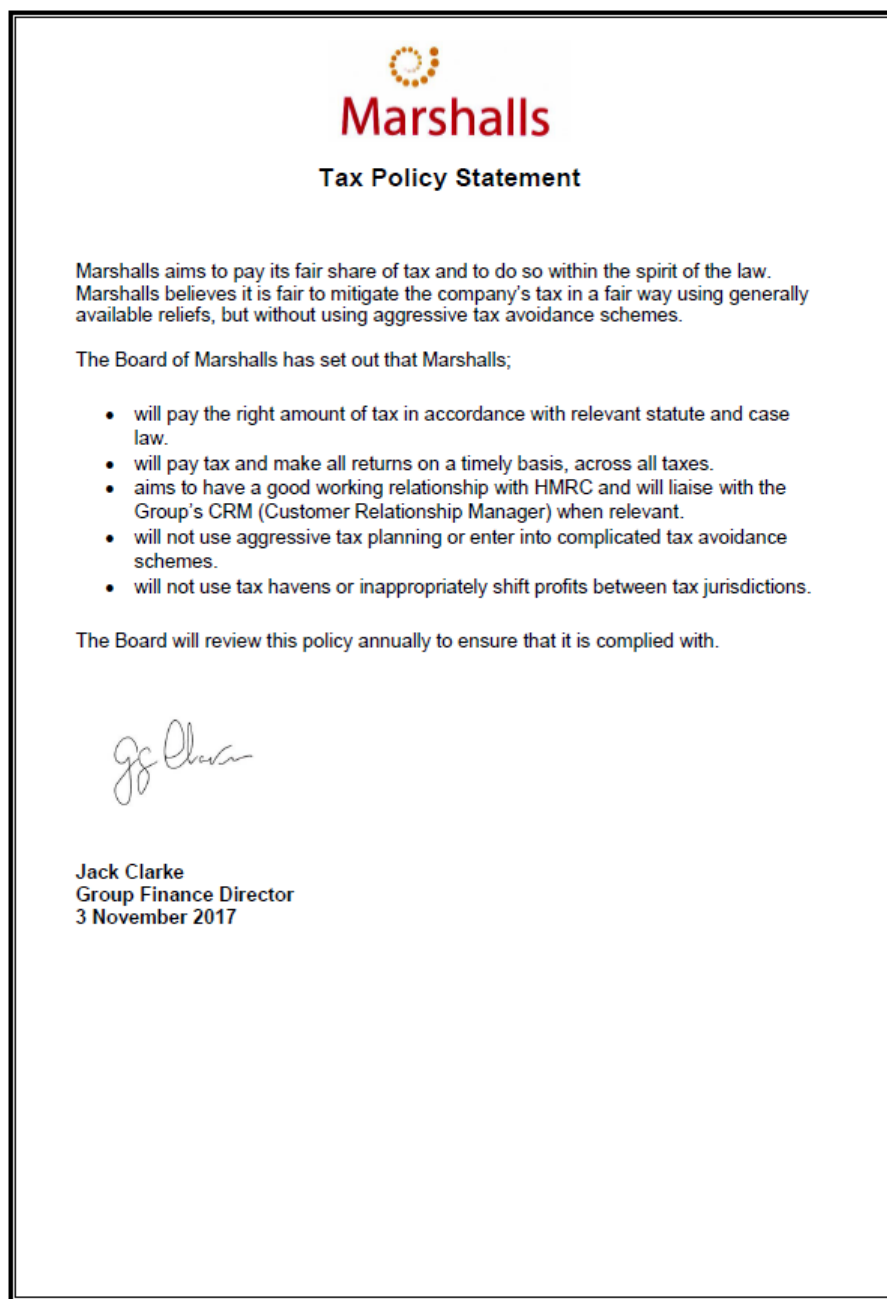
On behalf of the Board of Directors of Travis Perkins plc

Date: 06 November 2017

Appendix A: Examples of U.K. Tax Strategy Disclosures (continued)

Example Two

Below is the initial tax strategy disclosure by Marshalls PLC, a manufacturer of home improvement and landscaping products. Marshalls PLC is a FTSE 250 firm listed on the London Stock Exchange. The following disclosure can be accessed via the link below:
<https://www.marshalls.co.uk/documents/policies/taxpolicystatement.pdf>



Appendix B: Variable Definitions

Variable Name	Definition
<u>Difference-in-differences Variables</u>	
<i>Capex</i>	Capital expenditures [CAPX] scaled by total assets [AT]. This variable is reset to zero if the numerator is missing.
<i>CETR</i>	One-year cash effective tax rate, calculated as cash taxes paid [TXPD] scaled by pretax income before special items [PI - SPI]. Special items are reset to zero if missing. This variable is reset to missing if the denominator is non-positive, and its value is winsorized at zero and one.
<i>ETR</i>	One-year book effective tax rate, calculated as income tax expense [TXT] scaled by pretax income before special items. Special items are reset to zero if missing. This variable is reset to missing if the denominator is non-positive, and its value is winsorized at zero and one.
<i>Intangible</i>	Intangibles [INTAN] scaled by total assets. This variable is reset to zero if the numerator is missing.
<i>Inventory</i>	Inventory [INVT] scaled by total assets. This variable is reset to zero if the numerator is missing.
<i>Leverage</i>	Long-term debt [DLTT] scaled by total assets. This variable is reset to zero if the numerator is missing.
<i>Loss</i>	Indicator variable set equal to one if a firm reports pretax loss before special items in any of the past three years (i.e., $t-3$ to $t-1$), and zero otherwise.
<i>Post</i>	Indicator variable set equal to one for fiscal years beginning after September 15, 2016, when the U.K. tax strategy disclosure requirement became effective, and zero otherwise.
<i>PPE</i>	Property, plant, and equipment [PPENT] scaled by total assets. This variable is reset to zero if the numerator is missing.
<i>R&D</i>	Research and development expense [XRD] scaled by total assets. This variable is reset to zero if the numerator is missing.
<i>ROA</i>	Pretax income [PI] scaled by total assets.
<i>Size</i>	Natural log of one plus total assets in U.S. dollars. Where necessary, I convert foreign currencies to U.S. dollars using the exchange rate on the last day of the month in which the firm's fiscal year ends. Exchange rates are obtained from the Federal Reserve Bank and OANDA website.
<i>Treat</i>	Indicator variable set equal to one if the firm meets at least one disclosure threshold, and zero otherwise.
<u>Cross-sectional Variables</u>	
<i>Brand</i>	Indicator variable set equal to one if a firm owns at least one brand in the 2017 U.K. Top 50 or 2018 U.K. Top 75 most valuable brands list by Kantar Millward Brown, and zero otherwise. <i>Source: Kantar Millward Brown</i>
<i>Brand_House</i>	Indicator variable set equal to one if a firm owns a valuable brand and adopts a "houses of brands" strategy (Austin and Wilson 2017), and zero otherwise.

Appendix B: Variable Definitions (continued)

Cross-sectional Variables (continued)

<i>Brand_Other</i>	Indicator variable set equal to one if a firm owns a valuable brand and adopts either a “corporate branding” or a “mixed branding” strategy (Austin and Wilson 2017), and zero otherwise.
<i>Haven_Sub</i>	Indicator variable set equal to one if a firm has at least one subsidiary in a tax haven country, and zero otherwise. <i>Souce: Bureau van Dijk’s Orbis</i>

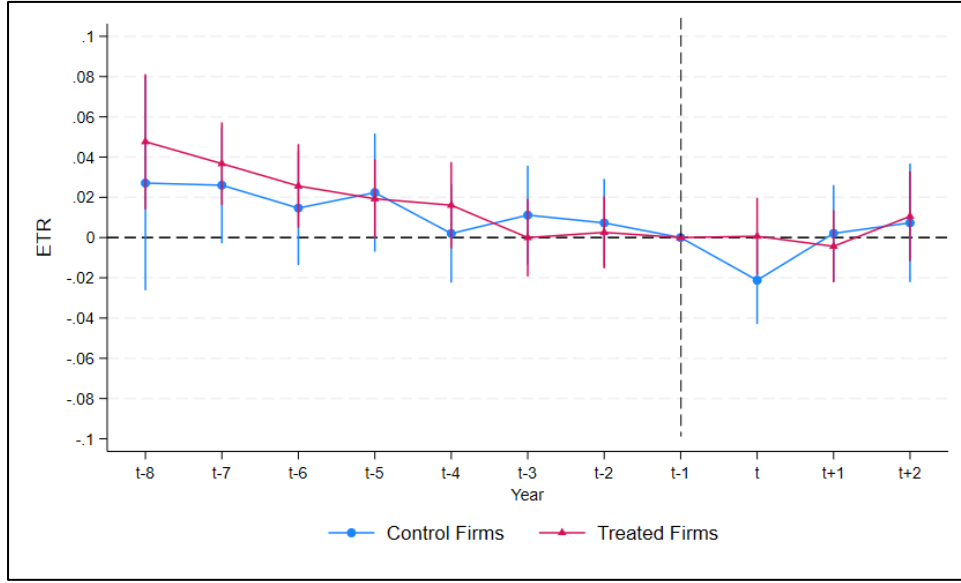
Other Variables

<i>NoDisc_RiskAppt</i>	Indicator variable set equal to one if a firm discloses tax risk appetite in its U.K. tax strategy disclosure, and zero otherwise. <i>Source: Hand-collected data</i>
<i>RV_Turnover</i>	The difference between the firm’s turnover in the year immediately prior to the regulation becoming effective and the £200 million disclosure threshold.

All variables are constructed using the Compustat Global database unless stated otherwise. Data items are listed in square brackets. All continuous variables are winsorized at the 1st and 99th percentile unless stated otherwise.

Figure 1: Tax Avoidance by Year

Panel A: *ETR as the measure of tax avoidance*



Panel B: *CETR as the measure of tax avoidance*

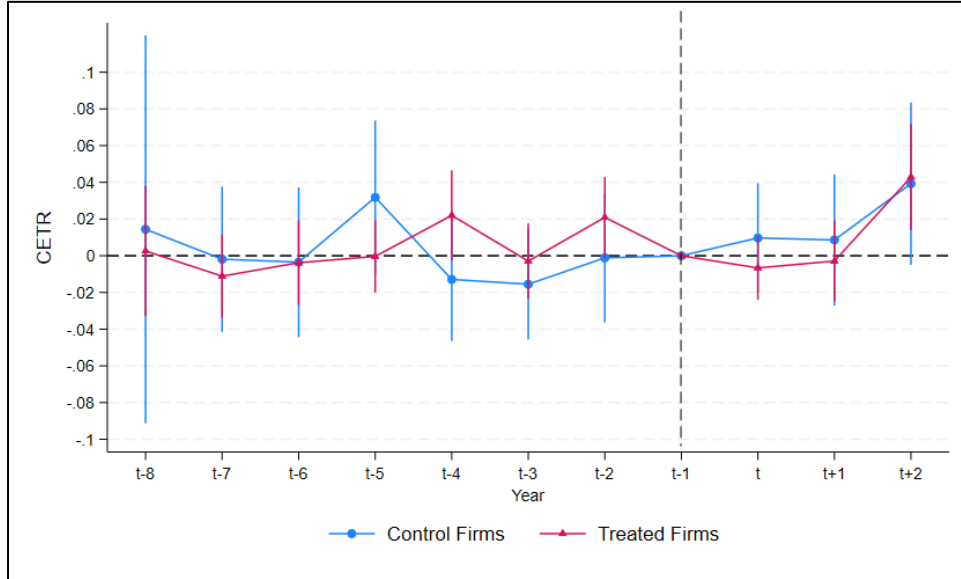


Figure 1 plots the level of tax avoidance by year separately for treated and control firms. In Panel A (Panel B), the measure of tax avoidance is *ETR* (*CETR*). In both panels, year *t-1* serves as the benchmark and represents the year immediately prior to the U.K. tax strategy disclosure requirement becoming effective. The vertical axis represents the incremental amount of *ETR* and *CETR* in a specific year to the value in year *t-1*. Appendix B provides detailed variable definitions.

Table 1: Sample Selection

	# Obs.	# Firms
U.K.-incorporated firms listed on the London Stock Exchange from 2010 to 2019	17,987	2,334
Less:		
Mutual funds	(2,161)	(264)
Member firms within the same group	(245)	(29)
Missing control variables	(1,564)	(87)
Missing <i>ETR</i>	(5,544)	(517)
Not present in both the pre- and post-regulation periods	(1,553)	(573)
Sample using <i>ETR</i>	6,920	864
Sample using <i>CETR</i>	5,080	624

Table 1 describes detailed sample selection procedures to create the sample used to estimate equation (1) under the difference-in-differences design. The sample size for the cross-sectional tests is reduced and varies based on the availability of the cross-sectional variables.

Table 2: Descriptive Statistics – DiD Design**Panel A: Full sample**

Variable	N	Mean	StdDev	Min	P25	P50	P75	Max
<i>Treat</i>	6,920	0.520	0.500	0.000	0.000	1.000	1.000	1.000
<i>Post</i>	6,920	0.303	0.459	0.000	0.000	0.000	1.000	1.000
<i>ETR</i>	6,920	0.192	0.168	0.000	0.086	0.188	0.250	1.000
<i>CETR</i>	5,080	0.214	0.182	0.000	0.108	0.188	0.265	1.000
<i>Size</i>	6,920	6.173	2.398	1.577	4.440	5.969	7.718	13.266
<i>Leverage</i>	6,920	0.139	0.161	0.000	0.000	0.084	0.227	0.692
<i>Intangible</i>	6,920	0.219	0.233	0.000	0.010	0.127	0.384	0.814
<i>Inventory</i>	6,920	0.086	0.140	0.000	0.000	0.019	0.122	0.761
<i>R&D</i>	6,920	0.013	0.032	0.000	0.000	0.000	0.007	0.174
<i>PPE</i>	6,920	0.187	0.231	0.000	0.016	0.086	0.279	0.899
<i>Capex</i>	6,920	0.030	0.039	0.000	0.002	0.015	0.041	0.194
<i>ROA</i>	6,920	0.082	0.088	-0.137	0.028	0.063	0.116	0.445
<i>Loss</i>	6,920	0.188	0.391	0.000	0.000	0.000	0.000	1.000

Panel B: Treated vs. control firms

Variable	<i>Treat = 0</i>			<i>Treat = 1</i>			Mean Diff.
	N	Mean	P50	N	Mean	P50	
<i>Post</i>	3,320	0.311	0.000	3,600	0.295	0.000	0.016
<i>ETR</i>	3,320	0.179	0.171	3,600	0.204	0.198	-0.025 ***
<i>CETR</i>	2,190	0.206	0.169	2,890	0.221	0.196	-0.015 ***
<i>Size</i>	3,320	4.354	4.395	3,600	7.854	7.593	-3.501 ***
<i>Leverage</i>	3,320	0.094	0.019	3,600	0.181	0.152	-0.087 ***
<i>Intangible</i>	3,320	0.218	0.112	3,600	0.219	0.138	-0.001
<i>Inventory</i>	3,320	0.074	0.008	3,600	0.097	0.031	-0.023 ***
<i>R&D</i>	3,320	0.017	0.000	3,600	0.010	0.000	0.007 ***
<i>PPE</i>	3,320	0.158	0.051	3,600	0.214	0.130	-0.056 ***
<i>Capex</i>	3,320	0.026	0.011	3,600	0.033	0.022	-0.007 ***
<i>ROA</i>	3,320	0.088	0.066	3,600	0.077	0.062	0.011 ***
<i>Loss</i>	3,320	0.269	0.000	3,600	0.113	0.000	0.156 ***

Table 2 presents descriptive statistics for the sample used to estimate equation (1). Panel A provides descriptive statistics for the full sample. Panel B provides select descriptive statistics separately for treated (*Treat* = 1) and control (*Treat* = 0) firms and reports tests of differences in the mean. All continuous variables are winsorized at the 1st and the 99th percentiles. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 3: Changes in Tax Avoidance – DiD Design

Panel A: Primary regression results

	Pred.	Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post</i> × <i>Treat</i>	+	0.005 (0.54)	0.004 (0.42)	-0.014 (-1.09)	-0.014 (-1.07)
<i>Size</i>	+/-		0.002 (0.25)		0.004 (0.31)
<i>Leverage</i>	+/-		0.018 (0.51)		0.002 (0.05)
<i>Intangible</i>	-		0.002 (0.07)		0.010 (0.19)
<i>Inventory</i>	-		0.086 (0.83)		-0.098 (-0.81)
<i>R&D</i>	-		-0.157 (-1.14)		-0.199 (-0.94)
<i>PPE</i>	-		0.123** (2.00)		0.059 (0.81)
<i>Capex</i>	-		-0.166 (-1.19)		0.103 (0.67)
<i>ROA</i>	+		0.082 (1.30)		-0.617*** (-8.12)
<i>Loss</i>	+/-		0.015 (1.46)		-0.017 (-1.12)
N.		6,920	6,920	5,080	5,080
Adjusted R ²		0.293	0.295	0.285	0.321
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 3, Panel A presents the results of estimating equation (1). *Post* is an indicator variable set equal to one for years after the U.K. tax strategy disclosure requirement became effective. *Treat* is an indicator variable set equal to one for firms that are subject to the U.K. tax strategy disclosure requirement. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 3: Changes in Tax Avoidance – DiD Design (continued)

Panel B: Parallel trend assumption

	Pred.	Dependent Variable =	
		<i>ETR</i>	<i>CETR</i>
		(1)	(2)
<i>Year_t-8 × Treat</i>	+/-	0.020 (0.64)	0.002 (0.04)
<i>Year_t-7 × Treat</i>	+/-	0.012 (0.65)	-0.001 (-0.06)
<i>Year_t-6 × Treat</i>	+/-	0.012 (0.65)	0.007 (0.30)
<i>Year_t-5 × Treat</i>	+/-	-0.003 (-0.16)	-0.025 (-1.10)
<i>Year_t-4 × Treat</i>	+/-	0.015 (0.92)	0.035* (1.73)
<i>Year_t-3 × Treat</i>	+/-	-0.010 (-0.61)	0.013 (0.74)
<i>Year_t-2 × Treat</i>	+/-	-0.004 (-0.29)	0.024 (1.20)
<i>Year_t × Treat</i>	+	0.021 (1.46)	-0.011 (-0.66)
<i>Year_t+1 × Treat</i>	+	-0.007 (-0.43)	-0.009 (-0.45)
<i>Year_t+2 × Treat</i>	+	0.003 (0.14)	0.008 (0.31)
Control Variables		Included	Included
N.		6,920	5,080
Adjusted R ²		0.295	0.322
Fixed Effects		Firm, Year	Firm, Year
Cluster		Firm	Firm

Table 3, Panel B presents the results of estimating equation (1) after replacing *Post* with indicators for individual years over the sample period. *Year_t-1* serves as the benchmark and represents the year immediately prior to the U.K. tax strategy disclosure requirement becoming effective. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 4: Cross-Sectional Tests Based on Ownership of Valuable Brands

Panel A: Primary regression results

	Pred.	Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post</i> × <i>Brand</i>	+	-0.025 (-1.13)	-0.020 (-0.90)	-0.008 (-0.33)	-0.014 (-0.61)
<i>Size</i>	+/-		0.006 (0.56)		0.030** (2.44)
<i>Leverage</i>	+/-		0.049 (1.05)		0.056 (1.14)
<i>Intangible</i>	-		0.001 (0.02)		-0.034 (-0.54)
<i>Inventory</i>	-		0.173 (1.50)		0.013 (0.09)
<i>R&D</i>	-		0.074 (0.23)		-0.069 (-0.16)
<i>PPE</i>	-		0.118 (1.42)		0.036 (0.35)
<i>Capex</i>	-		-0.125 (-0.55)		0.128 (0.45)
<i>ROA</i>	+		0.245*** (2.61)		-0.552*** (-5.69)
<i>Loss</i>	+/-		0.029* (1.78)		-0.001 (-0.02)
N.		3,600	3,600	2,890	2,890
Adjusted R ²		0.282	0.289	0.319	0.359
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 4, Panel A presents the results of estimating equation (2) within the subsample of firms that are subject to the U.K. tax strategy disclosure (*Treat* = 1). The cross-sectional variable is *Brand*, which captures firms' ownership of (or lack thereof) valuable brands. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 4: Cross-Sectional Tests based on Ownership of Valuable Brands (continued)

Panel B: Decomposing branding strategy

	Pred.	Dependent Variable =			
		ETR		CETR	
		(1)	(2)	(3)	(4)
<i>Post × Brand_House</i>	+	0.009 (0.11)	0.017 (0.20)	-0.068 (-1.01)	-0.067 (-0.97)
<i>Post × Brand_Other</i>	+	-0.030 (-1.28)	-0.024 (-1.07)	0.003 (0.10)	-0.005 (-0.21)
<i>Size</i>	+/-		0.006 (0.55)		0.030** (2.44)
<i>Leverage</i>	+/-		0.048 (1.03)		0.057 (1.15)
<i>Intangible</i>	-		0.002 (0.04)		-0.036 (-0.57)
<i>Inventory</i>	-		0.173 (1.50)		0.014 (0.09)
<i>R&D</i>	-		0.081 (0.25)		-0.084 (-0.19)
<i>PPE</i>	-		0.121 (1.47)		0.030 (0.30)
<i>Capex</i>	-		-0.125 (-0.54)		0.129 (0.46)
<i>ROA</i>	+		0.245*** (2.61)		-0.551*** (-5.68)
<i>Loss</i>	+/-		0.029* (1.77)		-0.000 (-0.02)
N.		3,600	3,600	2,890	2,890
Adjusted R ²		0.282	0.289	0.319	0.359
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm
<i>Post × Brand_House vs. Post × Brand_Other</i>					
<i>F-stat</i>		0.20	0.23	0.99	0.75
<i>p-value</i>		0.656	0.629	0.321	0.387

Table 4, Panel B presents the results of estimating equation (2) within the subsample of firms that are subject to the U.K. tax strategy disclosure (*Treat* = 1). The cross-sectional variables capture firms that adopt 1) a “house of brands” strategy (*Brand_House*) and those with 2) a “corporate branding” strategy or a mixed branding strategy (*Brand_Other*). All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 5: Cross-Sectional Tests Based on Tax Haven Operations

	Pred.	Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post</i> × <i>Haven_Sub</i>	+	-0.006 (-0.51)	-0.007 (-0.56)	-0.014 (-0.91)	-0.020 (-1.28)
<i>Size</i>	+/-		0.014 (0.91)		0.023* (1.68)
<i>Leverage</i>	+/-		0.010 (0.23)		0.007 (0.14)
<i>Intangible</i>	-		-0.029 (-0.49)		-0.052 (-0.78)
<i>Inventory</i>	-		0.143 (1.32)		0.043 (0.33)
<i>R&D</i>	-		-0.486 (-1.48)		-0.017 (-0.03)
<i>PPE</i>	-		0.087 (1.24)		0.088 (1.00)
<i>Capex</i>	-		-0.007 (-0.04)		0.226 (0.79)
<i>ROA</i>	+		0.358*** (3.93)		-0.548*** (-5.11)
<i>Loss</i>	+/-		0.022 (1.34)		-0.010 (-0.54)
N.		2,764	2,764	2,267	2,267
Adjusted R ²		0.324	0.339	0.317	0.353
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 5 presents the results of estimating equation (2) within the subsample of firms that are subject to the U.K. tax strategy disclosure (*Treat* = 1). The cross-sectional variable is *Haven_Sub*, which is set equal to one if the firm has at least one subsidiary in a tax haven country. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 6: Changes in Tax Avoidance – Regression Discontinuity Design

	Dependent Variable =			
	<i>ETR</i>		<i>CETR</i>	
	(1)	(2)	(3)	(4)
RD Estimate	0.005 (0.11)	-0.059 (-1.08)	-0.008 (-0.16)	-0.085 (-1.23)
Original Sample N.	6,920	6,920	5,080	5,080
RD Sample N.	2,074	2,074	1,518	1,518
Bandwidth (in £Millions)	172	262	260	365
Total Obs. in Bandwidth	872	1,340	910	945
Polynomial Order	Linear	Second-Order	Linear	Second-Order
Kernel	Triangular	Triangular	Triangular	Triangular
Include Controls	No	No	No	No
Include Fixed Effects	No	No	No	No

Table 6 presents the results of estimating the effect of the U.K. tax strategy disclosure requirement on firms' tax avoidance using a regression discontinuity (RD) design. The sample for this test is restricted to observations in the post-regulation period. Columns (1) and (3) estimate a nonparametric local linear model, and columns (2) and (4) estimate a second-order polynomial model. The RD estimate is reported in the first row, and the *t*-statistics calculated using robust bias-corrected standard errors are reported in parentheses below the RD estimate. All specifications use the optimal bandwidth that minimizes the mean-squared error and a triangular kernel function. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 7: Changes in Tax Avoidance – DiD Design with U.K. Domestic Firms

		Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post</i> × <i>Treat</i>	+	0.010 (0.57)	0.008 (0.41)	0.014 (0.66)	0.017 (0.86)
<i>Size</i>	+/-		-0.023 (-1.08)		-0.040 (-1.35)
<i>Leverage</i>	+/-		0.028 (0.46)		-0.096 (-1.03)
<i>Intangible</i>	-		0.000 (0.00)		0.199* (1.95)
<i>Inventory</i>	-		0.188 (0.83)		0.289*** (3.08)
<i>R&D</i>	-		-0.541* (-1.81)		-1.070*** (-4.64)
<i>PPE</i>	-		0.020 (0.16)		0.043 (0.32)
<i>Capex</i>	-		-0.041 (-0.13)		0.082 (0.25)
<i>ROA</i>	+		0.030 (0.25)		-0.164 (-1.63)
<i>Loss</i>	+/-		0.029 (1.21)		-0.079*** (-2.86)
N.		1,172	1,172	634	634
Adjusted R ²		0.221	0.225	0.303	0.353
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 7 presents the results of estimating equation (1) using a subsample of U.K. domestic firms. *Post* is an indicator variable set equal to one for years after the U.K. tax strategy disclosure requirement became effective. *Treat* is an indicator variable set equal to one for firms that are subject to the U.K. tax strategy disclosure requirement. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 8: Excluding Firms Subject to the Country-by-Country Reporting Requirement

	Pred.	Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post × Treat</i>	+	0.018 (1.55)	0.017 (1.44)	-0.014 (-0.84)	-0.012 (-0.71)
<i>Size</i>	+/-		0.001 (0.13)		-0.006 (-0.40)
<i>Leverage</i>	+/-		0.032 (0.83)		0.027 (0.54)
<i>Intangible</i>	-		0.019 (0.47)		0.041 (0.66)
<i>Inventory</i>	-		0.048 (0.39)		-0.259 (-1.55)
<i>R&D</i>	-		-0.256* (-1.85)		-0.273 (-1.29)
<i>PPE</i>	-		0.112 (1.53)		0.033 (0.36)
<i>Capex</i>	-		-0.242 (-1.54)		0.012 (0.07)
<i>ROA</i>	+		0.003 (0.05)		-0.611*** (-6.64)
<i>Loss</i>	+/-		0.012 (1.06)		-0.020 (-1.09)
N.		5,138	5,138	3,572	3,572
Adjusted R ²		0.298	0.299	0.270	0.305
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 8 presents results of estimating equation (1) after excluding firms that are subject to the Country-by-Country (CbC) reporting requirement from the sample. All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.

Table 9: Cross-Sectional Tests based on Risk Appetite Disclosure

	Pred.	Dependent Variable =			
		<i>ETR</i>		<i>CETR</i>	
		(1)	(2)	(3)	(4)
<i>Post</i> × <i>NoDisc_RiskAppt</i>	+	-0.016 (-1.09)	-0.017 (-1.17)	-0.017 (-0.84)	-0.010 (-0.58)
<i>Size</i>	+/-		-0.014 (-0.96)		0.031** (2.43)
<i>Leverage</i>	+/-		0.093* (1.78)		-0.012 (-0.24)
<i>Intangible</i>	-		0.054 (0.63)		-0.055 (-0.61)
<i>Inventory</i>	-		0.126 (0.96)		0.206** (2.18)
<i>R&D</i>	-		-0.572 (-1.50)		-0.806** (-2.50)
<i>PPE</i>	-		0.078 (0.93)		0.021 (0.24)
<i>Capex</i>	-		0.655* (1.91)		0.614* (1.91)
<i>ROA</i>	+		0.302** (2.56)		-0.564*** (-4.38)
<i>Loss</i>	+		0.039* (1.78)		0.022 (0.71)
N.		1,873	1,873	1,482	1,482
Adjusted R ²		0.248	0.266	0.341	0.392
Fixed Effects		Firm, Year	Firm, Year	Firm, Year	Firm, Year
Cluster		Firm	Firm	Firm	Firm

Table 9 presents results of estimating equation (2) within the subsample of firms that are subject to the U.K. tax strategy disclosure (*Treat* = 1). The cross-sectional variable captures the absence of a firm's tax risk appetite disclosure (*NoDisc_RiskAppt*). All continuous variables are winsorized at the 1st and the 99th percentiles. Firm and year fixed effects are included, and standard errors are clustered by firm. *t-stats* are reported in parentheses below the corresponding coefficient estimate. ***, **, * indicates statistical significance at the one, five, and ten percent level (two-tailed), respectively. Appendix B provides detailed variable definitions.